

Beyond being one of the greatest artists ever to have lived, Leonardo da Vinci was a brilliant Renaissance thinker whose work spanned engineering, science, architecture, and mathematics. He believed that observation was a vital tool for inquiry and that art and science were equally important fields of creative knowledge.

MILESTONES

EARLIEST WORK

Paints *The Annunciation* in 1473 while apprenticed to Verrocchio. It is his earliest surviving work.

MOVES TO MILAN

Designs military and engineering projects for the Duke of Milan from 1483.

MILITARY ENGINEER

Works for Cesare Borgia in Rome from 1502, surveying and drawing highly accurate maps.

PERSONAL COLLECTION

Devotes more time to sketching and writing in notebooks following the death of his father in 1504.

ROYAL INVITATION

Invited to the royal court by King François I of France in c.1516; dies there in 1519.

Leonardo da Vinci was born near the town of Vinci in Tuscany, Italy, in 1452. He was educated as the son of a gentleman despite being the illegitimate son of a Florentine legal notary, Piero, and Caterina, a peasant woman. As a teenager, he became the apprentice of Verrocchio, one of the foremost artists of Florence, and learned painting, sculpting, and metalworking, along with mathematics and the sciences.

Possessing a remarkably artistic eye, da Vinci also had an insatiable curiosity for science and mechanics. At the age of 17, he assisted Verrocchio in the construction of a 2-ton (1.8-tonne) gilded copper sphere, which was installed on the summit of Florence cathedral's 376-ft- (114-m-) high dome in 1469. He was fascinated by the mechanical working of the hoists used to lift the sphere to such a height.

Human mechanics

In keeping with the artistic education of the Renaissance period, da Vinci received a grounding in anatomy during his apprenticeship and studied the skeletal and muscular structure of the human figure. As revealed by the contents of his notebooks—many of which were

As well as practical, applied schemes for engineering, architecture, and physiology, da Vinci's wealth of sketches include a design (dating c.1490) for a flying machine, which he named an "ornithopter."

